Ordering Analysis of Gene Expression Dynamics

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Abstract

DNA microarrays technology can be used to discover the expression of thousands of genes and the regulatory relationship between them. This article presents an ordering method to analyze gene expression dynamics, and find the ordering relationship, such as the upstream or downstream between these genes. Our ordering method has two components: we use an ordering index (OI) combined with standard statistic significant test to analyze gene expression data. We use ordering index to measure the association of genes through the dynamics of gene expression. Because the characteristics of our ordering index are same to the 「conditional probability」, which make the model can present the direction of gene controlling. The hypothesis direction can provide us more information about the regulatory mechanisms of genes. And the standard statistic significant test can make the judgments objective.

1 Introduction

The human genomic project has provided us with a lot of data and biological knowledge, and the development of cDNA microarrays technology has given scientists more opportunities to study physiology and gene function in depth. One of the advantages of this technology is that enable investigators to simultaneously measure the expression of thousands of genes, and find the gene expression patterns that control our body. When analyzing the different gene expression levels, we can do systematical analysis about the main function of the genes and the cross relationship of between them through this new technology.

The technique of cDNA Microarrays allows us to analyze a great number of genes at a time. This is a very powerful tool when analyzing the whole genome. There are a lot of methods developed to analyze the gene expression data that obtained from cDNA Microarrays experiments. According the complex levels, we can roughly separate them into about two levels.

First, by using the differences of gene expression level, we can decide the gene expression level if the statistical significance can be qualified on the cell situation in this time. Typically, the standard statistic procedure can be used. The statistical t-test is the most useful procedure, and the t-statistic results can be changed to become a score to evaluate the significance [1]. There is another method that can get more background message using the prior knowledge through the Bayesian statistics [2].

Secondly, a more complex way is used to analyze the mutual relationship between genes or a series of genes. Use the gene expression pattern and the level of gene expression to analyze the relationship between the genes that we are interesting. Methods using in this level are more complex than the first level. A common method uses the Pearson correlation coefficient to construct a gene similarity matrix, and then it can use the hierarchical clustering algorithm based on the average-linkage method to do the classification of the genes [3, 4].

On the other hand, the classification based on the Pearson correlation coefficient are easily influential by the data that obtained from the experiments, such as the outlier or the changing condition of experiment resulted the data can not present on the normal level. We can treat each experiment data as a random variable. To prevent the research bias because of the random variable, we can choose the entropy as the measurement of the gene expression level, because entropy does not influenced by the value of random variable which is the measurement of the uncertainty of the distribution. Algorithm using the maximum entropy can analyze the genome-wide expression patterns [5]. To construct relevance networks using the pair-wise mutual information of the gene can present the relationship more clearly [6, 7]. This method set up threshold mutual information (TMI=1.3) by experience to judge the associations of the genes exists or not. Although to set up a TMI and to construct a relevance network can be more easily to present the mutual relationship of the genes, but it also has a disadvantage. That is, this method can not detect the association with the mutual information lower than the threshold.

Our method uses the「ordering index」 (OI) which comes from the procedure that using mutual information divides to the marginal entropy, and replaces 「mutual information」 to measure the association of genes through the gene expression data. The result presents the aspect of the gene association. Because the characteristics of our ordering index are the same to the 「conditional probability」,which make the model having the ability to present the controlling direction of the genes. Because of our ordering index has the range between 0 to 1, and there are an exact relation between mutual information and the correlation coefficient derived from two variables [8]. So we can approximate change the value to the t-statistic because of OI is similar to the one-way correlation coefficient. Then use the right-tail hypothesis test to the one-way correlation relationship and determine if the OI is significant enough to say this pair of the genes existing ordering relationship.

The hypothesis direction can provide more information about the gene expression dynamics. And the statistic significant test can make the judgments more objective than the threshold TMI determined by subjective experience. Here, we use this hypothesis association to present the ordering relationship of the genes. Therefore, we can use the direction of OI to represent the controlling direction of the genes.

2 METHOD

First calculated the entropy of each gene and the pair-wise mutual information of the gene pair. Then we calculated OI of each gene pair as the measurement of the ordering relationship.

Secondly, the use of the statistical test replaces the critical value to judge OI, and take the pairs of genes with the statistical significance to bind the association network of genes. Because of the direction characteristic containing in 「ordering index」, somehow the result gets from the model has the ability to present the controlling direction of the genes.

Uniform distribution has the maximum entropy, so if the entropy of the gene expression data is higher, and it means that its expression levels are more randomly distributed.

In general, the entropy of gene A, $H(A)$, is defined by equation (1)

$$H(A) = -\sum_{i=1}^{m} p(a_i)\log_2 p(a_i) \qquad (1)$$

By calculating the entropy as the measurement of gene expression level, we use a histogram technique [6] to calculate the range of value for each gene expression data, divide the range into 5 sub-interval (m=5), and then count the relative frequency $p(a_i)$ of data which lie in each sub-interval a_i.

The joint entropy of the two genes A and B, $H(A,B)$, is defined by equation (2)

$$H(A,B) = -\sum_{i=1}^{m}\sum_{j=1}^{m} p(a_i,b_j)\log_2 p(a_i,b_j) \qquad (2)$$

Here the $p(a_i,b_j)$ represents the relative frequency which lies in the cell (a_i, b_j). One advantage of using entropy is that we can present the variation of gene expression level, and it does not need to consider the question about normalization, so that it will not meet the question about suitable normalization equation.

Using the entropy to calculate the mutual information of pair-wise gene, as shown in equation (3)

$$MI(A,B) = H(A) + H(B) - H(A,B) \qquad (3)$$

The higher mutual information shows the higher association between two genes. Because mutual information represents the association, so we choose MI(A, B) to represent the association between genes. Although the value of mutual information will not be a negative value ($MI(A,B) \geq 0$), the value still could be higher than 1, which is not an ideal measurement. Considering the property of $MI(A,B) \leq H(B)$, we can extend the definition to the ordering index, $OI(A/B)$, as shown in equation (4)

$$OI(A/B) = \frac{MI(A,B)}{H(B)} \qquad (4)$$

Because of our ordering index follow the form of the「conditional probability」, so we can see the OI has the same characteristics with 「conditional probability」 which make our model can present ordering relationship between data. And we use the ordering relationship to represent the direction of gene controlling. The hypothesis direction can provide us more information about the regulatory mechanisms of gene. Ordering index restricts the direction of regulatory association of the genes, so we use OI(A/B) to represent the ordering relationship from gene A to gene B. OI(A/B) is always not equal to OI(B/A). It means the relation can not be reversible. There is another useful property of OI(A/B), i.e. $0 \leq \mathrm{OI}\ (A/B) \leq 1$.

The range of OI(A/B) is approximate to the correlation coefficient. We can use statistical hypothesis test to judge the significant of the ordering relationship, i.e. OI(A/B) [9]. Statistical hypothesis test provide us more objective information than the methods that using the critical value to be a threshold. Statistical hypothesis test can provide the estimation of false positive rate and false negative rate. Because of the value of OI(A/B) just between 0 to 1, so we can change OI(A/B) to t-statistic than use the right-tail t-test to assess the ordering relationship between gene A and gene B.

$$H_0 : r \leq 0 \quad vs \quad H_1 : r > 0 \quad at\ a = 0.05/0.01$$

$$t_{AB} = \frac{OI(A/B)}{\sqrt{\frac{1-[OI(A/B)]^2}{n-2}}} \sim t(n-2) \qquad (5)$$

If $t_{AB} > t_{1-a}(n-2)$, it represents the existence of the association from A to B, where n = the number of sample.

In this paper, we use the hypothesis test of correlation coefficient to carry out the test of OI, because it approximates the correlation coefficient in which the degree of freedom is (n-2). Through the above null hypothesis H_0 and the t-statistics, we can analyze the association of genes, and in theory, we can control the error rate under the level as we expect, through the choice of the value of α.

3 Experimental data set

We use the public data set obtained from the Stanford Microarray database SMD [10]. We obtain the stomach cancer data set, which we draw out 17 p53-related genes as a small data set, and each gene has 20 samples. And the entire gene name refers from the ExPASy Molecular Biology Server [11]. We use these genes to test the ability of the explanation and the reasonability of our model.

This small data set contains normal and tumor samples, which come from 20 patients with each gene expression level. The genes that we choose are shown in Table I.

Table I

Gene Symbol		Gene Name
TP53BP1	G1	tumor protein p53 binding protein, 1
P5326	G2	hypothetical protein p5326
NM23-H6	G3	nucleoside diphosphate kinase type 6 (inhibitor of p53-induced apoptosis-alpha)
TP53TG3	G4	TP53TG3 protein
PIGPC1	G5	p53-induced protein PIGPC1
TP53INP1	G6	tumor protein p53 inducible nuclear protein 1
RRM2B	G7	ribonucleotide reductase M2 B (TP53 inducible)
MDM2	G8	**Mdm2, transformed 3T3 cell double minute 2, p53 binding protein (mouse)
TP53BP2	G9	**tumor protein p53 binding protein, 2
TP53	G10	tumor protein p53 (Li-Fraumeni syndrome)
TP53TG1	G11	TP53 target gene 1
REPRIMO	G12	candidate mediator of the p53-dependent G2 arrest
PP5395	G13	hypothetical protein PP5395
PIG11	G14	**p53-induced protein
DDA3	G15	p53-regulated DDA3
TP53BPL	G16	tumor protein p53-binding protein
PA26	G17	p53 regulated PA26 nuclear protein

All these show in Table I. are p53-related genes, which contain the upstream and downstream genes of the p53.

4 RESULTS

We use the method with α=0.05 and α=0.01 to analyze the association of these genes, and we obtain the relevance networks with hypothesis direction of the gene controlling direction. The value of α means the probability which we reject the hypothesis wrongly. When the value of α is chosen, we choose the probably error rate of the association when we consider this association is true.

In another word, we choose the false positive rate that we can bear. Hence, when we choose α = 0.01, this means we just want to bear 1% false positive rate.

We calculate both directions of OI, which

Table II.

OI (A/ B) →

α=0.05	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15	G16	G17
G1		-	-	-	-	-	*	-	*	-	-	-	*	-	-	-	-
G2	-		*	*	-	-	-	-	-	-	-	-	-	-	-	-	-
G3	-	*		-	-	-	-	-	-	-	-	-	-	-	-	*	-
G4	-	*	-		-	-	-	-	*	*	-	-	*	*	*	-	-
G5	-	-	-	-		-	*	-	-	-	-	*	*	-	-	-	*
G6	-	-	-	-	-		-	-	*	-	-	-	-	-	-	-	-
G7	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-	-
G8	-	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-
G9	*	-	-	-	-	-	*	-		*	-	-	*	-	-	-	-
G10	-	-	-	-	-	-	-	-	*		-	-	-	-	-	-	-
G11	-	-	-	-	-	-	-	-	-	-		-	-	-	-	-	-
G12	-	-	-	-	*	-	-	-	-	*	-		-	*	-	-	-
G13	*	-	-	*	*	-	-	-	*	-	-	-		-	*	-	-
G14	-	-	-	-	-	-	-	-	-	-	-	*	-		*	-	-
G15	-	-	-	*	-	-	-	-	-	-	-	-	-	*		*	-
G16	-	-	*	-	-	-	-	-	-	-	-	-	-	-	*		-
G17	-	-	-	-	*	-	-	-	-	*	-	-	-	-	-	-	

← OI (B/ A)

express by OI (A/B) and OI (B/A), and we can get pair-wise OI as we show below.

Table II presents the association in which OI is significant enough when we take the value of α = 0.05. The direction of OI(A/B) contains 21 pairs of gene which is significant enough. As shown in Table II, we can see that these pairs of gene have relationship and the relationship has the direction A->B. Another direction OI(B/A) contains 18 pairs of genes which is significant enough and have the relationship with direction B->A. Fourteen of these have both direction significance. We use the symbol "*"to represent the pair of gene with significant OI under α = 0.05.

We use the OI as the measurement of the association. This means we assume that the result of our experiment can give hypothesis direction of the gene. If the pair of the gene only has one-way association indicated these two genes may have the upstream and downstream relationship. This direction obeys the direction of OI. If both directions of the pair of genes are significance that may means these two genes interaction with each other or that may have another gene interaction with these two genes in the pathway. Next we will use the network that find on α = 0.01 to analyze above assumption. Table III presents the association in which OI is significant enough when we take the value of α = 0.01. The direction of OI(A/B) contains 5 pairs of gene which are significant enough. We can see these pairs of gene have relationship and the relationship has the direction A->B. Another direction OI(B/A) contains 6 pairs of genes which are significant enough and have the relationship with direction B->A. 2 of these have both direction significant. We use the symbol "**"to represent the pair of gene with the significance of OI under α = 0.01. We can use Figure I to present the frequencies of association when the value of α = 0.05 and α = 0.01.

Figure I

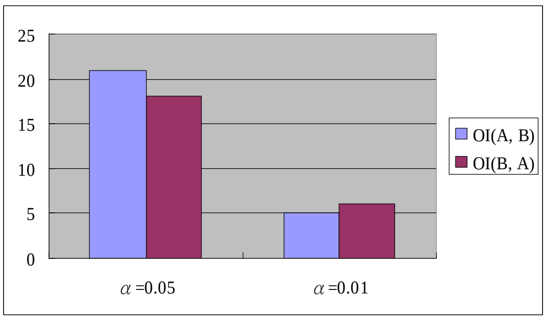


Table III.

OI (A/ B) →

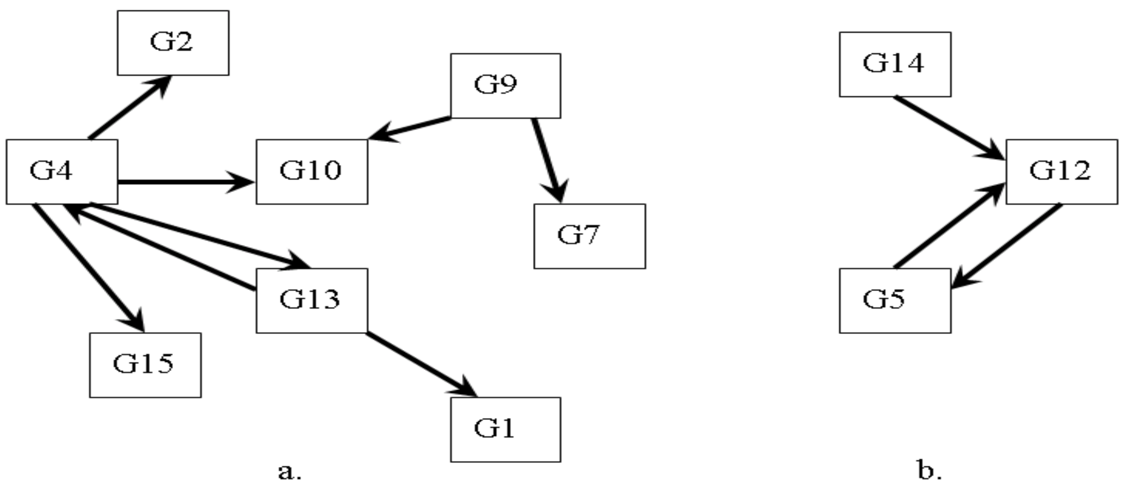
←OI (B/ A)

α=0.01	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	G11	G12	G13	G14	G15	G16	G17
G1		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
G2	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
G3	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-
G4	-	**	-		-	-	-	-	-	**	-	-	**	-	**	-	-
G5	-	-	-	-		-	-	-	-	-	-	**	-	-	-	-	-
G6	-	-	-	-	-		-	-	-	-	-	-	-	-	-	-	-
G7	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-	-
G8	-	-	-	-	-	-	-		-	-	-	-	-	-	-	-	-
G9	-	-	-	-	-	-	**	-		**	-	-	-	-	-	-	-
G10	-	-	-	-	-	-	-	-	-		-	-	-	-	-	-	-
G11	-	-	-	-	-	-	-	-	-	-		-	-	-	-	-	-
G12	-	-	-	-	**	-	-	-	-	-	-		-	-	-	-	-
G13	**	-	-	**	-	-	-	-	-	-	-	-		-	-	-	-
G14	-	-	-	-	-	-	-	-	-	-	-	**	-		-	-	-
G15	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-	-
G16	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		-
G17	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	

Figure I shows when we reduced the false positive rate we can bear (α = 0.05 change to α = 0.01), we can find the less number of pair that is significant enough. If all genes in the data set have truly association, the value of α change may represent the distance of these associated genes. In fact, most of the association we found have lower MI than TMI = 1.3. However, these relationships seem reasonable. That means using our method can find the association of gene more objective, and this method may find more pairs of gene which have association but do not have particular high MI. Also it can prevent higher rate of false negative.

We present the relevant network of Table III in Figure II.

Figure II.



As shown in Figure II, we can see the Figure II present two networks of gene, Figure II (a) and Figure II (b). Figure II (a) contains 8 genes, and we can see their relationship very easily and clearly. G9 represents the tumor suppressor p53-binding protein 2 (TP53BP2), which function is a regulator that plays a central role in regulation of apoptosis and cell growth via its interactions, which regulates TP53 by enhancing the DNA binding and transactivation function of TP53 on the promoters, belonging to the upstream gene of P53. G10 represents the tumor protein p53 which belongs to the downstream gene of G9. G7 represents the ribonucleoside diphosphate reductase M2, whose function provides the precursors necessary for DNA synthesis, which also belongs to the downstream gene of G9.

We can see that the network based on our model is a reasonable network.

Beside two hypothetical proteins G2 and G13, the rest of these genes contain G4 and G4 represents the TP53-inducible gene (TP53 target gene 3) which is suspected that this protein plays an important role in the TP53-mediated signaling pathway [12]. This gene may have interaction with P53. G15 represents p53-regulated DDA3 and G1 represents tumor suppressor p53-binding protein 1. This network seems reasonable.

In Figure II (b), G14 represents p53-induced protein 11 which is apoptosis regulator. G12 represents candidate mediator of the p53-dependent G2 arrest which is cell-cycle control gene, G5 represents P53-induced protein PIGPC1, and all belonging to downstream genes of P53 should have interaction with each other.

5 DISCUSSIONS

There are two main advantages in our model. One is that we can present a relation network with suggestive direction. Even some of them are still blurred, but this suggestive direction can indicate the gene regulatory direction under this specific situation maybe some kind of disease or some specific cell condition that depending on the data we got. Thus we can get more information than other method, and we can understand the relation of gene regulatory network more clearly.

The other is that we can use the statistical test to evaluate the significant level of OI and this is a more objective method, because in this test we can control the false positive rate and can prevent increasing false negative rate.

We suggest there are several questions and potential development still waiting for solving. One of these is to set up an automatic analysis procedure, allow this model can analyze mass data automatically. This should be our short term work. When analyzing this model, we find the OI or t-statistics seem containing the distant meaning inside, and find a useful method to determine the relation distance which may be a potential work. Besides, the gene pair that contains two directions needs mass data to analyze to find the clear relationship. This may be an interesting work.

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